Project: Worked examples
Title: Section properties
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates area, centroid, second moments, moduli, radii of gyration, and torsion and warping constants of a parametric, polygon or composite section. The outline has x horizontal and y vertical; properties use EN 1993 axes, y-y horizontal and z-z vertical.

WHAT IT CALCULATES	RELATION
Area and centroid	$A = ½ \oint (x\,dy - y\,dx)$; $x_c = \int x\,dA / A$; $y_c = \int y\,dA / A$
Second moment about y-y	$I_y = \int y^2\,dA - A\,y_c^2$, by Green's theorem and the parallel-axis theorem
Second moment about z-z	$I_z = \int x^2\,dA - A\,x_c^2$; product $I_{yz} = \int xy\,dA - A\,x_c\,y_c$
Principal axes	$\tan 2\theta = -2\,I_{yz} / (I_y - I_z)$
Elastic section modulus	$W_{el,y} = I_y / y_{max}$
Plastic section modulus about the equal-area axis, found by bisection	$W_{pl,y} = \lvert Q_{above}\rvert + \lvert Q_{below}\rvert$
Torsion constant J	Saint-Venant series (solid rectangle), $\pi d^4/32$ (circle), Bredt (closed thin-walled), $\Sigma bt^3/3$ (open)
Warping constant of an I-section	$I_w = I_z\,(h - t_f)^2/4$

Basis: Properties of sections; torsion of solid and thin-walled sections[1](Table A.1 and Table 10.1, 8th ed.); Torsional properties of rolled I-shapes[2](App., 1st ed.); Hot rolled steel channels, I and H sections: dimensions and masses[3]; Hot finished structural hollow sections: tolerances, dimensions and sectional properties[4].

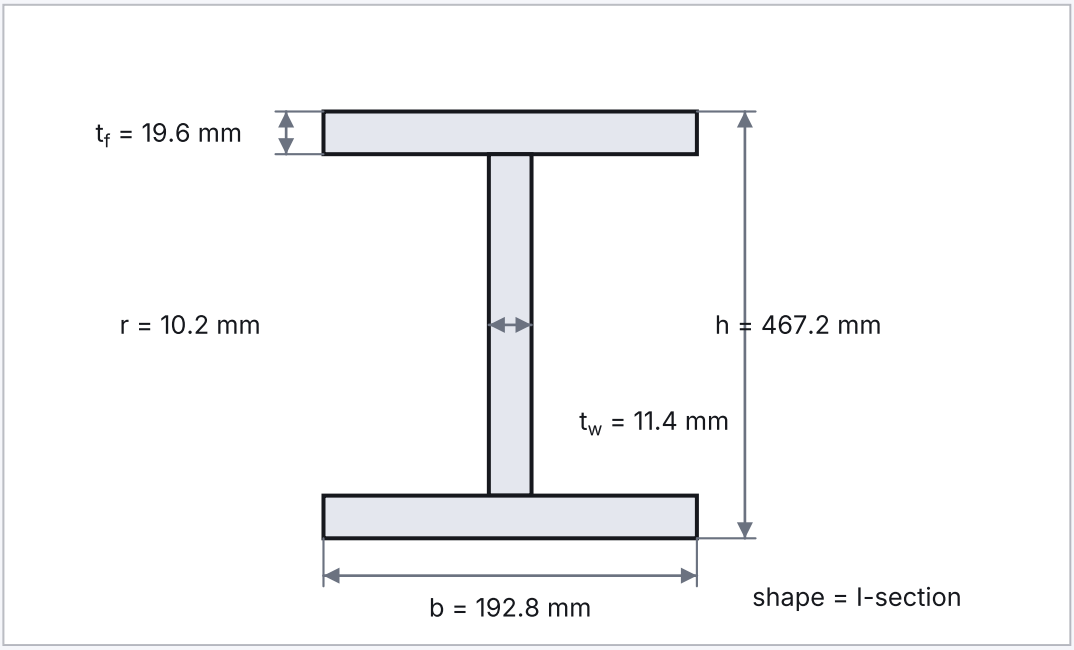


Figure 1 — A doubly symmetric I-section b by h, web t, flanges t_f, root radius r. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
shape	Shape	I-section		spec: BS EN 10365 UB 457×191×98	any
h	Depth or vertical leg	467.2	mm	spec: BS EN 10365 UB 457×191×98	1 … 5000 mm
b	Width or horizontal leg	192.8	mm	spec: BS EN 10365 UB 457×191×98	1 … 5000 mm
t_w	Wall, web or leg thickness	11.4	mm	spec: BS EN 10365 UB 457×191×98	0.1 … 500 mm
t_f	Flange thickness	19.6	mm	spec: BS EN 10365 UB 457×191×98	0.1 … 500 mm
r	Root radius of an I-section, channel, tee or angle, or outer corner radius of a hollow rectangle (optional; blank = 0 for the rolled shapes, 1.5 t_w for a hollow rectangle)	10.2	mm	spec: BS EN 10365 UB 457×191×98	0 … 500 mm

3. Assumptions

A1. A polygon outline does not cross itself. (Green's theorem integrates the signed loops as given.)
A2. Holes lie inside the outline and do not overlap one another. (Green's theorem integrates the signed loops as given.)
A3. Rolled-section root fillets are quarter circles of radius r and flange tips are square. (Published section tables use the same idealisation.)
A4. Hollow-section corners are quarter circles of radius 1.5 t_w outside and 1.0 t_w inside by default. (BS EN 10210-2 hot-finished idealisation.)
A5. Thin-walled torsion constants neglect the junction and tip effects except the fillet term kept for rolled I-sections. (Open sections: $\Sigma bt^3/3$; closed: Bredt.)

4. Working

$[7]\ A = \tfrac{1}{2} \oint (x\,dy - y\,dx)$ [1](Table A.1)

= Green's theorem over 108 vertices in 1 loop(s)

= 125.27 cm²

Area

$[8]\ x_c = \int x\,dA / A$

= Green's theorem over 108 vertices in 1 loop(s)

= 0 mm

Centroid x from the shape origin

$[9]\ y_c = \int y\,dA / A$

= Green's theorem over 108 vertices in 1 loop(s)

= 0 mm

Centroid y from the shape origin

$[10]\ I_y = \int y^2\,dA - A\,y_c^2$

= Green's theorem over 108 vertices in 1 loop(s)

= 45728 cm⁴

Second moment of area about the centroidal y-y axis

$[11]\ I_z = \int x^2\,dA - A\,x_c^2$

= Green's theorem over 108 vertices in 1 loop(s)

= 2347 cm⁴

Second moment of area about the centroidal z-z axis

[12] $I_{yz} = \int xy\ dA - A\ x_c\ y_c$

= Green's theorem over 108 vertices in 1 loop(s)

= 0 cm⁴

Product of inertia about the centroidal y-y and z-z axes

[13] $I_1 = (I_y + I_z)/2 + \text{sqrt}(((I_y - I_z)/2)^2 + I_{yz}^2)$

= (45728 cm⁴ + 2347 cm⁴) / 2 + sqrt(((45728 cm⁴ - 2347 cm⁴) / 2)^2 + (0 cm⁴)^2)

= 45728 cm⁴

Major principal second moment

[14] $I_2 = (I_y + I_z)/2 - \text{sqrt}(((I_y - I_z)/2)^2 + I_{yz}^2)$

= (45728 cm⁴ + 2347 cm⁴) / 2 - sqrt(((45728 cm⁴ - 2347 cm⁴) / 2)^2 + (0 cm⁴)^2)

= 2347 cm⁴

Minor principal second moment

[15] $\theta = ½\ \text{atan2}(-2\ I_{yz},\ I_y - I_z)$

= ½ atan2(0, 43381)

= 0 deg

Angle from the y-y axis to the major principal axis, anticlockwise

[16] $y_{top} = y_{max} - y_c$

= Green's theorem over 108 vertices in 1 loop(s)

= 233.6 mm

Extreme fibre above the centroid

[17] $y_{bot} = y_c - y_{min}$

= Green's theorem over 108 vertices in 1 loop(s)

= 233.6 mm

Extreme fibre below the centroid

[18] $x_{ext} = \max(x_{max} - x_c,\ x_c - x_{min})$

= Green's theorem over 108 vertices in 1 loop(s)

= 96.4 mm

Extreme fibre from the centroid, x

[19] $W_{el,y} = I_y\ /\ \max(y_{top},\ y_{bot})$

= 45728 cm⁴ / max(233.6 mm, 233.6 mm)

= 1957.5 cm³

Elastic section modulus about the y-y axis, governing fibre

[20] $W_{el,z} = I_z\ /\ x_{ext}$

= 2347 cm⁴ / 96.4 mm

= 243.47 cm³

Elastic section modulus about the z-z axis

[21] $W_{pl,y} = |Q_{above}| + |Q_{below}|$ about the equal-area axis

= equal-area axis at y = 0 mm

= 2232.5 cm³

Plastic section modulus about the y-y axis

[22] $W_{pl,z} = |Q_{right}| + |Q_{left}|$ about the equal-area axis

= equal-area axis at x = 0 mm

= 378.9 cm³

Plastic section modulus about the z-z axis

[23] $r_{g,y} = \text{sqrt}(I_y\ /\ A)$

= sqrt(45728 cm⁴ / 125.27 cm²)

= 191.06 mm

Radius of gyration about the y-y axis

[24] $r_{g,z} = \text{sqrt}(I_z\ /\ A)$

= sqrt(2347 cm⁴ / 125.27 cm²)

= 43.286 mm

Radius of gyration about the z-z axis

[25] $\alpha_{pl,y} = W_{pl,y} / W_{el,y}$
$= 2232.5\ cm^3 / 1957.5\ cm^3$
$= 1.1404$
Shape factor about the y-y axis
[26] $J = (2\ b\ t_f^3 + (h - 2\ t_f)\ t_w^3)/3 + 2\ \alpha\ D^4 - 0.42\ t_f^4$
$= b = 192.8,\ t_f = 19.6,\ h = 467.2,\ t_w = 11.4,\ r = 10.2\ mm$
$= 121.29\ cm^4$
Torsion constant
[27] $I_w = I_z\ (h - t_f)^2 / 4$ [2](DG9)
$= 2347\ cm^4 \times (467.2 - 19.6)^2\ mm^2 / 4$
$= 1175500\ cm^6$
Warping constant

5. Results

SYMBOL	VALUE	DESCRIPTION
A	125.27 cm²	Area
I_y	45728 cm⁴	Second moment of area about the y-y axis
I_z	2347.0 cm⁴	Second moment of area about the z-z axis
I_{yz}	0 cm⁴	Product of inertia about the y-y and z-z axes
$W_{el,y}$	1957.5 cm³	Elastic section modulus about the y-y axis
$W_{el,z}$	243.5 cm³	Elastic section modulus about the z-z axis
$W_{pl,y}$	2232.5 cm³	Plastic section modulus about the y-y axis
$W_{pl,z}$	378.9 cm³	Plastic section modulus about the z-z axis
$r_{g,y}$	191.1 mm	Radius of gyration about the y-y axis
$r_{g,z}$	43.29 mm	Radius of gyration about the z-z axis
J	121.3 cm⁴	Torsion constant
I_w	1175539 cm⁶	Warping constant
x_c	0 mm	Centroid x
y_c	0 mm	Centroid y
I_1	45728 cm⁴	Major principal second moment
I_2	2347.0 cm⁴	Minor principal second moment
θ	0°	Principal axis angle
$\alpha_{pl,y}$	1.1404	Shape factor about the y-y axis

6. Checks (governing first)

None.

7. Applicability

WARNING J: the open thin-walled sum with the rolled-section fillet term is used. [thin-walled-torsion]

8. Verification

Template struct.section-properties@1.2.0 Content hash a95bd0059585bd36 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Seaburg PA, Carter CJ. Torsional Analysis of Structural Steel Members. Steel Design Guide 9. American Institute of Steel Construction; 1997.
3. British Standards Institution. Hot Rolled Steel Channels, I and H Sections—Dimensions and Masses. BS EN 10365. BSI.
4. British Standards Institution. Hot Finished Steel Structural Hollow Sections—Part 2: Tolerances, Dimensions and Sectional Properties. BS EN 10210-2. BSI.